Reinforcement Focus+Context

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Abstract – A technique for graph-based visualisations is introduced we nick-name Reinforcement Focus+Context (RFC), which is useful to navigate in graphs of complex structure like generated content networks or ontologies. RFC lays out prominent nodes and their surroundings as "structural primitives", such as trees or circles that are embedded in the complete graph. This way, special aspects of the graph content become much more intelligible in focal regions.

Keywords – visualisation, focus+context, fisheye, graph layout, graph filtering, hyperbolic distortion

1. Introduction

Many structures at the basis of artificial intelligence and specifically intelligent information retrieval (IIR) applications are graphs, and often such graph structures are directly relevant for user examination and interaction: Graph edges often provide the viewer with intuitive understanding of the relational structure of the data, supporting human associative thinking.

However, direct use of "node–link" graph visualisation is difficult when it comes to larger datasets, and often graph visualisations have been considered impractical as a means of user presentation and interaction: (1) Space-limited visual displays of graphs become cluttered and difficult to interpret if the size of the graph ($\sim$ number of vertices) and/or its connectivity ($\sim$ number of edges) increases, and (2) the graph visualisation performance does not usually scale well to larger data volume, regarding both computational load and layout quality; only few drawing algorithms can claim to be able to automatically layout graphs in the range of $10^{3..4}$ nodes [7]. Further, the size of graphs strongly influences the way how to interact with the visualisations, and less intelligibiligy of large graph's details impedes interaction.

These problems have lead to alternative types of visualisation for graph-based data, e.g., by providing hyperlink text pages as in the Citeseer system [4] or by multiple lists of vertices as in the author–topic browsing system by Steyvers et al. [10]. For a more detailed presentation of local graph structure, these "local browsing" approaches sacrifice the direct view on the relational structure beyond adjacent vertices.

Research on the inherent problems of graph visualisations proper (and corresponding interaction techniques) has lead to a number approaches for manipulating the visualisation, a good survey of which was provided by Herman et al. [7]. For the present article, such manipulation approaches can be categorised as (1) structural and (2) geometrical approaches. Structural approaches scale the structural detail of the graph by discrete operations, e.g., clustering subgraphs, whereas geometric approaches scale the size of presentation.

This scaling is controlled by interaction techniques, which are outlined in Table 1. A common problem with the "trivial" interactions zooming and panning is that they detach the visible subgraph from the rest of the data, leading to a loss of context and therefore a usability problem (similar to the above browsing examples).

The most widely used method to reduce this decontextualisation problem is the focus+context (FC) paradigm, according to which data of high estimated interest are presented in one or more emphasised focal regions while a downscaled view of the surrounding context is shown in par-

interaction	structure	geometry
zoom	show↔hide	scale viewport
pan	step focus	move viewport
FC	rel. aggregation	scale focus
RFC	—	reinforce focus

Table 1. Structural and geometric manipulation.

allel. The FC paradigm originates from Furnas' fisheye views[1] [3], and the "classical" FC example was contributed by Lamping et al. with the hyperbolic browser [8]. This geometric FC technique embeds a tree within a hyperbolic space, resulting in a fisheye-like layout that maps the infinite Euclidean space to a disc with high geometric emphasis at its center.

A structural analogon to the geometric FC approach is to perform structural aggregation of nodes deemed less important[7], depending on the estimated interest of the user. This was also already integrated in Furnas' general concept [3].

As a third approach to an FC concept, this article introduces reinforcement focus+context (RFC). Rather than emphasising a focal region by geometric magnification or structural alteration, reinforcement focus+context reinforces the layout of the region to appear as "structural primitives": structures such as trees or circles formed by the relevant subgraphs that can be easily recognised and understood by a human viewer even while embedded in the context of a complexly laid out graph.

2. Reinforcement focus+context

Reinforcement focus+context follows the idea that a human viewer recognises relatively simple structures within a more complex context in a similar way as when looking at a magnified structure. This follows the finding that some visual aesthetics, for instance a reduction of edge crossings, in a graph visualisation are strong facilitators of intelligibility [9, 11].

[1]Focus+context, fisheye and distortion-oriented presentation are used rather synonymously in the literature.

2.1. Suitable subgraphs

A first task in designing an RFC method is to find simple structures that are likely to conform to the graph morphology common to a particular visualisation task. In fact, the choice of subgraph to be reinforced could be lead by re-occurring subgraph structures that have semantic importance such as co-citation patterns in a bibliographic reference network, and in such a setting it would be desirable to have an automatic algorithm that recognises these structures as a result of structural clustering and/or user interaction.

To demonstrate the principle in this paper, we take a simpler approach and choose a tree-like subgraph (tree with additional edges possible). For our purpose, this has three benefits: Trees are guaranteed to exist as subgraphs (ignoring sibling linkage), the layout algorithm is relatively straight-forward, and such a visualisation of "focal trees" has can be thought of as analogous to a "mind map" [1], a pattern with a central topic and associated aspects commonly used to informally communicate structured information between people.

2.2. Layout algorithm

The RFC approach can be applied to graph layout algorithms generally. However, in the present case we will define the layout modification for a force-directed layout because this was used when developing the application. This layout algorithm is based on Fruchterman and Reingold's (FR) heuristic spring embedder [2] and has – with a couple of enhancements – shown good layout results for directed and undirected graphs with continuous animation.

The extended FR heuristic basically ensures the following constraints on every node pair:

- Hold adjacent vertices at a specific distance (optionally edge-type specific), tradeoff this distance using Hooke's law.
- Repel vertices by their center point with a force reciprocally proportional to the

squared distance (optionally by an elliptical shape, as proposed by Harel and Koren [5]).

- Limit the displacement of nodes at every iteration according to an annealing schedule to quickly stabilise the graph.

The task now is to augment these conditions to layout the subgraph around a center node as a tree, i.e., to *reinforce* local layout. For our illustrative tree example, this is straight-forward and can be achieved by increasing the RF edge spring constant and enforcing an angular constraint between neighboring pairs (e_i, e_j) of the N edges incident to the center/root: In principle, deviation from a homogeneous angular distribution results in a momentum on neighbouring edges:

$$M(e_i, e_j) = k_\alpha \Big[\angle(e_i, e_j) - \frac{2\pi}{N} \Big], \qquad (1)$$

where $\angle(e_i, e_j)$ is the angular diplacement between the edges, and k_α is the angular "spring constant", which is chosen to balance the effects of the FR layout algorithm and the local reinforced layout.

In the practical algorithm, such a momentum can be approximated by a longitudinal force between pairs of neighbouring branch nodes (n_i, n_j) that takes into account the distances between branch nodes and center node:

$$F(n_i, n_j) = k_r \Big\{ D(n_i, n_j) \quad - \big[D(n_0, n_i) + D(n_0, n_j) \big] \sin \frac{\pi}{N} \Big\}, \qquad (2)$$

where $D(a, b) = \sqrt{\lambda_x (a_x - b_x)^2 + \lambda_y (a_y - b_y)^2}$ is the Euclidean distance with weighted coordinates to facilitate layout of horizontally spreading text labels, and k_r a reinforcement spring constant.

Naturally, layout cannot be better than the angular resolution, which – for unweighted $D(\cdot, \cdot)$ – is bounded by the number of nodes N adjacent to the center, $R_\alpha \leq 2\pi/N$. For moderate node degrees ($2 < N < 15$) results appear good, and as a solution for more difficult situations, the hybrid approach below is suggested, i.e., a combination of layout, geometrical emphasis and structural FC.

In order to make sure that the area of the reinforced layout is uncluttered by non-focal nodes, an additional repelling force similar to the one in the FR algorithm can be introduced. For the tree-shaped reinforcement primitive shown here, the shape of the RFC area is elliptical, suggesting a force between non-focal vertices and the ellipse embedding the focus are similar to [5].

The algorithm can be extended to several foci, i.e., center nodes. Then a tradeoff between the reinforced layout trees is necessary. There are two cases: (1) If vertices are shared by the two focal trees, then for the corresponding edges a smaller $k'_\alpha \approx k_\alpha/2$ can prevent degenerate layout of the shared nodes. (2) If the centers are adjacent vertices, a trade-off between the elliptical regions is performed that repels all nodes not shared by the focal trees from the intersection region. Further, the focal–non-focal vertex force introduced in the last paragraph is not applied to any focal nodes.

Two figures give an illustrative example of the RFC approach. Consider the visualisation in Figure 1: Here an FR embedded graph is distorted by a hyperbolic mapping, which results in a radial shape of the focal region around the vertex "Project a"[2]. However, the nodes are laid out according to the FR forces, leading to an unevenly spaced tree around it center root[3]. Figure 2 shows the same graph, but uses the RFC approach. The case of two foci is visualised, "Project a" and "Project c". Note the effects of the conditions above: The focal tree branches are relatively evenly spaced. Within the intersection region (shaded ellipses), no node is accepted by a constraint. Further, focal–non-focal repelling force can be observed in the vertex "Group Alpha", which is displaced out of the center between "Person C" and "Person A".

[2] This typed, directed graph is composed of a network of content and social relations in a virtual community ("knowledge network"): People work in projects and create documents that can be used to identify expert knowledge. This follows the scenario in [6]. Moreover, people are aggregated by groups.

[3] The edge layout is not according to the hyperbolic distortion. Rather it has been found that a parabolic edge curving like the one shown increases intelligibility and aesthetics.

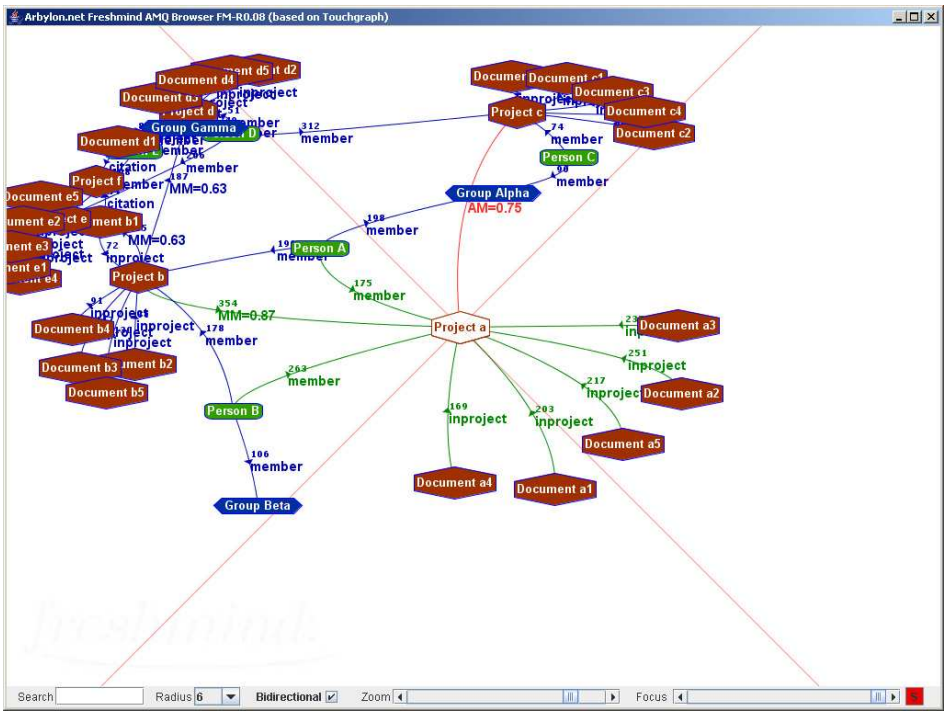


Figure 1. Geometrical focus+context in a hyperbolic graph view.

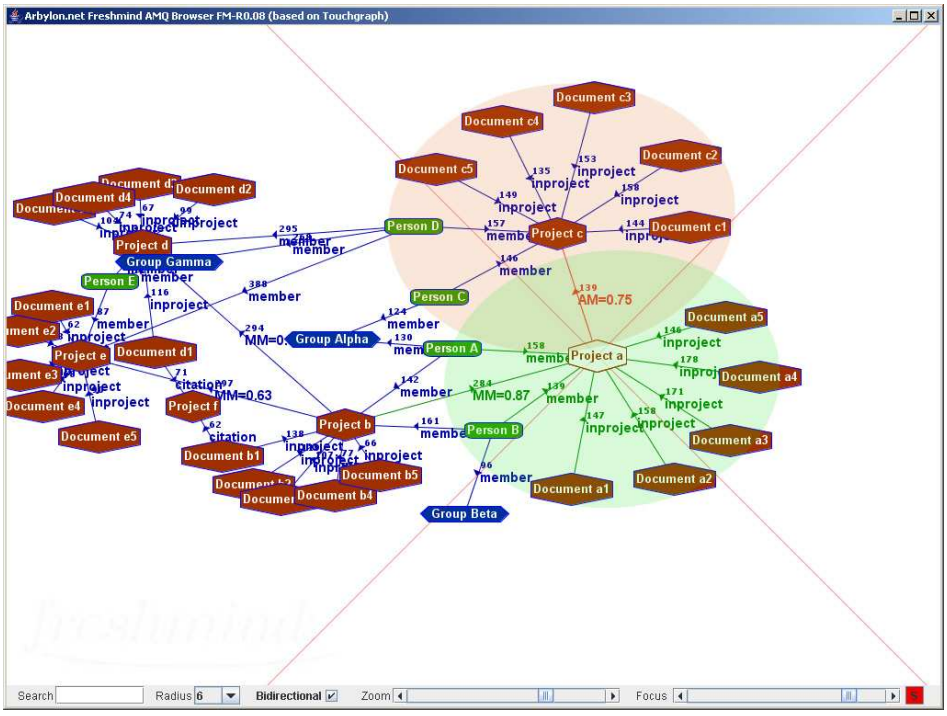


Figure 2. Reinforcement focus+context with two foci.

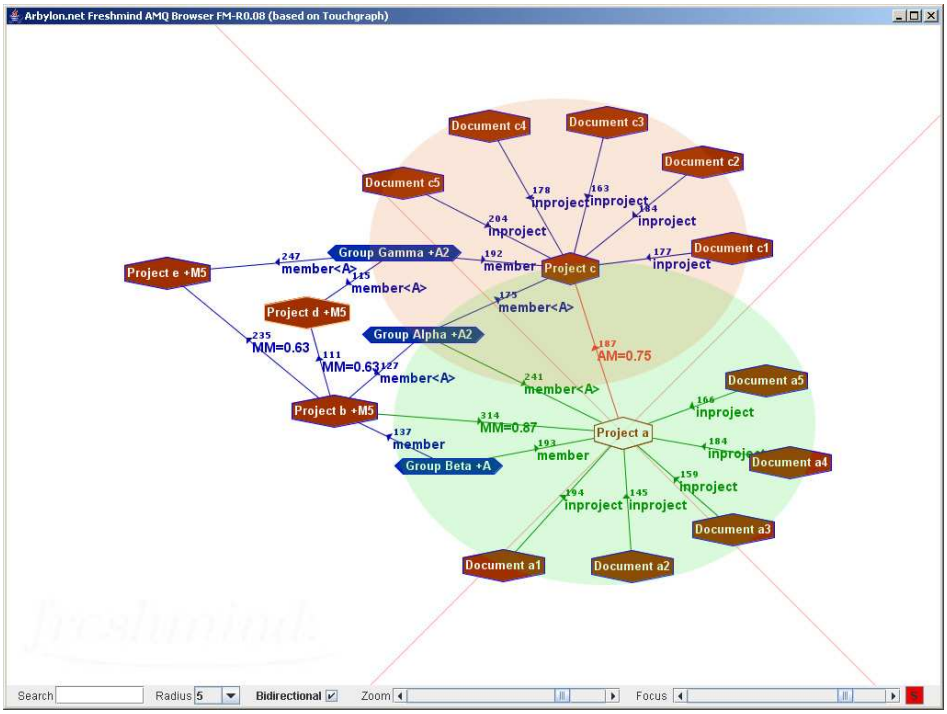


Figure 3. Hybrid focus+context visualisation using a combination of reinforcement and structural techniques.

2.3. Hybrid focus+context

In the typology introduced in this paper, RFC can be considered a geometrical approach as it does not work on the discrete graph structure but rather lays out the existing graph. This independence makes it possible to combine RFC with other focus+context methods. Laying out an enlarged tree structure as with geometric FC appears straight-forward: The two layout algorithms can be combined by first laying out the RFC structure and then applying the fisheye distortion with on the resulting coordinates.

Another example is shown in Fig. 3. Here a combination of RFC and structural FC is used. Compared to Fig. 2, one can see that, away from the focal region, node types with a lower level-of-detail (persons and documents) have been aggregated to their parent nodes, which is possible using predifined rules (e.g., leaves of local tree structures are hidden). Note that the vertex "Group Alpha" now has become "Group Alpha +A2" referring to the two aggregated persons[4]; further note that the vertex is directly laid out in the focal region of "Project a" and "Project c".

This hybrid approach is also expected useful to improve visualisations of more complex graphs, for instance by allowing to filter also certain types in the focal region.

Conclusions

We introduced the technique Reinforcement Focus+Context (RFC) for graph-based visualisations. RFC lays out prominent nodes and their surroundings as "structural primitives" that increase intelligibility of local regions of the graph content, which we demonstrated with a local tree-like structure. RFC can be considered an appropriate technique complementing existing Focus+Context and other interaction methods, where we have shown an example of a hybrid approach of structural and reinforcement focus+context techniques.

Future work will extend the idea towards other structural primitives and will look for possibilities to automatically recognise salient substructures in graphs. This is of special importance to apply RFC as a visual data mining technique.

[4] A in aggregation indicators like +A2 stands for *actors*, and M for *media*, generalisations of people and content in the graph data model used.

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